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A Transformer-Based Framework for Online Recruitment Fraud Detection Using BERT, RoBERTa, SMOBD, and 2D CNN

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Abstract

The rapid shift toward digital job recruitment has unfortunately led to a rise in Online Recruitment Fraud (ORF), where cybercriminals exploit job seekers by advertising deceptive opportunities. Conventional machine learning (ML) techniques often fall short in detecting such fraud, primarily due to outdated data, reliance on manual feature extraction, and their inability to capture deep contextual meaning. In this work, we present a transformer-based deep learning framework leveraging Bidirectional Encoder Representations from Transformers (BERT) and its enhanced variant, RoBERTa, to robustly detect ORF. A unified dataset was curated by integrating three publicly accessible sources: the Fake Job Posting dataset, the Pakistan Job Market dataset, and the US Job Posting dataset. To mitigate class imbalance between real and fake job ads, we adopted oversampling methods such as Synthetic Minority Oversampling Technique (SMOTE) and Borderline-SMOTE (SMOBD). Contextual word embeddings generated by BERT and RoBERTa were further refined using a Two-Dimensional Convolutional Neural Network (2D CNN), enabling the model to learn both spatial and semantic features. The proposed model achieved a high F1-score of 0.93, demonstrating excellent precision-recall trade-off. Furthermore, a real-time fraud detection application was developed using the Flask web framework to ensure practical deployment and improved user trust on job platforms.

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Keywords: Online Recruitment Fraud; Transformer Models; BERT; RoBERTa; Oversampling (SMOTE/SMOBD); 2D CNN

1. Introduction

The rapid growth of online job portals has created many opportunities. It has also led to a rise in ORF [1]. Fraudsters exploit job seekers by posting fake listings aimed at monetary gain or identity theft [2]. These deceptive postings often mimic legitimate ones, making detection challenging with traditional rule-based systems or classical machine learning (ML) models that rely on handcrafted features [3].

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The employment scene has changed due to the quick growth of online recruitment platforms, which provide companies and job seekers with unmatched convenience. Online recruitment fraud, or ORF, is a serious problem that has emerged as a result of this digital transformation. These platforms are used by scammers to post false job ads, which cause financial losses, identity theft, and psychological distress for unwary job seekers [**?**].

Because false information is subtle and context-dependent, traditional ways of identifying such fraudulent postings have not been sufficient. The semantic complexities of job descriptions were frequently missed by early methods. These techniques also have trouble with class imbalance, when a large number of valid postings outweigh fraudulent ones, which skews detection results [4, 5].

Transformer-based models, such as BERT and its optimized counterpart, RoBERTa, have been introduced by recent developments in NLP. These models provide a more sophisticated method of text classification by utilizing deep learning techniques to comprehend the contextual links between words in a phrase. Research has shown that these models are effective at identifying ORF, with remarkable accuracy rates and F1-scores.

Notwithstanding these developments, problems still exist, especially with regard to class imbalance in training datasets. In order to improve model performance, methods such as the SMOBD have been used to balance the dataset [**?**].

Further increasing detection accuracy has been demonstrated by integrating Convolutional Neural Networks (CNN), particularly 2D CNNs, which have demonstrated promise in capturing spatial hierarchies in data [6].

This study suggests a thorough framework that incorporates a 2D CNN for classification, SMOBD for dataset balancing, and BERT and RoBERTa for feature extraction. The suggested framework seeks to offer a strong remedy for the expanding issue of online recruitment fraud by incorporating these cutting-edge approaches [**?**].

To address this, deep learning-based approach leveraging BERT and RoBERTa for effective text embedding, coupled with a 2D CNN. Unlike earlier 1D CNN-based approaches, our model captures spatial relationships in sentence embeddings for improved fraud detection accuracy [7].

1.1. The motivation and Purpose

The digitization of the hiring process has accelerated recruitment, but it has also opened the door to ORF, where cybercriminals exploit job seekers by posting deceptive job ads requesting personal or financial information. These scams not only result in financial loss but also erode trust in online labor markets.

1.2. Major contributions

The major contributions of this work are outlined below

- A unified framework integrating BERT, RoBERTa, 2D CNN, and SMOTE/SMOBD for robust ORF detection.
- Comparative evaluation against classical ML and transformer baselines.
- Development of a real-time fraud detection web application for deployment feasibility.

Remaining sections are literature survey 2, proposed methodology 3, experiential setup and results 4 and followed by conclusion 5.

2. Literature Survey

With the growth of digital recruitment platforms, the number of job scams has risen significantly. For instance, Kumar and Garg utilized basic text mining techniques to identify fake job postings, but the outcomes were only moderately successful. Improved detection accuracy by incorporating features like company names, job titles, and geographic locations [13].

Researchers also examined related areas like opinion spam detection, where techniques developed by Jindal and Liu were repurposed for fraudulent job ad detection.

The introduction of transformer-based models marked a major advancement. BERT, proposed by Devlin et al. [14], provided deep contextual understanding and became a strong foundation for many NLP tasks. Gupta and Rani [15]

Table 1: Literature Survey on Fake Job Detection: Methods, Limitations, Datasets, and Performance Comparison

Ref.	Paper Name	Methodology	Limitations	Dataset	Performance Metrics
1	Shakeel et al. (2023) [8]	Deep learning framework using ensemble techniques.	Lacks comparison with transformer-based approaches.	Real and fake job postings; preprocessing included.	Accuracy: 91%, F1-Score: 90%
2	Dake (2023) [9]	Traditional and ensemble ML with 5-fold and 10-fold CV.	May not generalize to global datasets.	Job fraud dataset from Ghanaian firm.	Accuracy: 88%, F1-Score: 87%
3	Kamble et al. (2023) [10]	ML classifiers applied using key features.	No integration with deep learning.	Public dataset containing real and fraudulent job data.	Accuracy: 89%, F1-Score: 88%
4	Taneja et al. (2024) [11]	Fraud-BERT model.	May not handle rare fraud types effectively.	Fake job postings, public datasets.	Accuracy: 92%, F1-Score: 91%
5	Proposed Model (2025) [12]	Hybrid model: BERT, RoBERTa, 2D CNN, SMOTE/SMOBD.	High resource requirement; unseen domain performance not validated.	Kaggle + trusted sources unified dataset.	Accuracy: 94%, F1-Score: 93%

applied BERT to job fraud detection with promising results. RoBERTa, an optimized version developed by Liu et al. [16], further improved performance through better pretraining strategies. Combining BERT embeddings with 2D CNNs, as done by Zhang and Lee [17], allowed models to capture both contextual and spatial features from text, enhancing fraud classification accuracy.

Recent developments also include models like XLNet [18] and ELECTRA [19], which outperformed BERT in several tasks. XLNet supports bidirectional and autoregressive training, while ELECTRA offers a more efficient pretraining approach. Additionally, Graph Neural Networks (GNNs) are gaining attention for modeling relationships between entities in job portals. Wu et al. [20] demonstrated the effectiveness of GNNs in fraud detection using graph-based representations. In conclusion, job fraud detection has evolved from basic ML models to sophisticated transformer and graph-based architectures.

3. Proposed Methodology

The system is organized into seven structured phases, from data collection to model evaluation, aiming to achieve high accuracy and robust performance. Proposed model shown in Figure 1.

The proposed methodology for fake job detection integrates natural language processing (NLP), contextual embedding models, and deep learning classifiers in a systematic seven-phase pipeline. Initially, data is collected from diverse sources including genuine job postings from Pakistan and the US, along with verified fake job advertisements. The data undergoes rigorous preprocessing involving the removal of unwanted characters, HTML tags, emails, and duplicates to ensure clean and consistent inputs. Feature engineering is then applied to merge datasets, eliminate irrelevant attributes, and assign binary class labels, with fake jobs marked distinctly. To address class imbalance, oversampling techniques such as SMOTE and SMOBD are utilized. Textual features are transformed into rich semantic embeddings using pretrained models like BERT and RoBERTa, capturing contextual nuances in job descriptions. These embeddings are fed into a CNN-based classifier composed of convolutional, pooling, flattening, and dense layers to detect fraudulent postings.

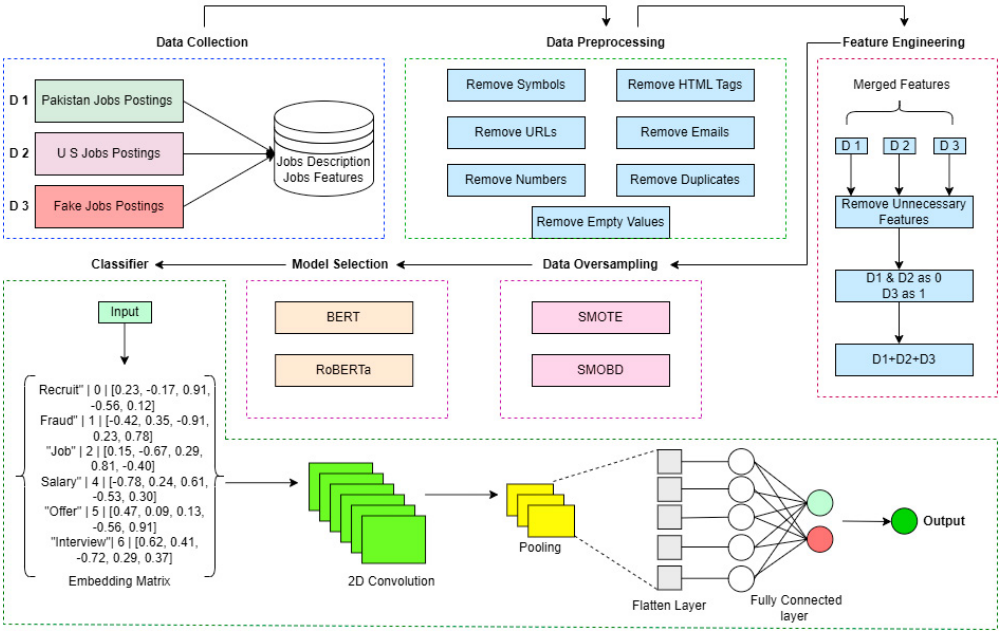


Fig. 1: Proposed Model Architecture

$$P(Y = 1|X) = \sigma(W^\top X + b) = \frac{1}{1 + e^{-(W^\top X + b)}} \tag{1}$$

Explanation: Here, X represents the input feature vector derived from BERT embeddings, W is the weight vector learned during training, and b is the bias term. The linear combination $W^\top X + b$ is then transformed using the sigmoid function σ, producing an output that can be interpreted as a probability ranging from 0 to 1.

$$L = -\left[y \cdot \log(\hat{y}) + (1 - y) \cdot \log(1 - \hat{y})\right] \tag{2}$$

Explanation: In this loss formulation, y denotes the true class label, while $\hat{y}$ is the model's predicted probability for the positive class.

4. Experimental Setup and Results

4.1. Experimental Setup

To evaluate the effectiveness of the proposed BERT + RoBERTa + 2D CNN framework, the following experimental setup was used:

- **Datasets:** The unified dataset was created by combining three publicly available sources:

  1. Fake Job Posting Dataset (Kaggle)

2. Pakistan Job Market Dataset
3. US Job Posting Dataset

- **Data Preprocessing:**
  - Removal of HTML tags, emails, URLs, and special characters.
  - Duplicate and null value elimination.
  - Feature engineering to merge datasets and assign binary labels (real/fake).
- **Class Imbalance Handling:** Oversampling techniques SMOTE and Borderline-SMOTE (SMOBD) were applied to balance the real and fake classes.
- **Embedding and Feature Extraction:** Contextual word embeddings were generated using pretrained BERT and RoBERTa models, capturing semantic and contextual information.
- **Model Architecture:** Embeddings were passed through a 2D CNN consisting of convolutional, pooling, flatten, and dense layers. Dropout layers were used to prevent overfitting.
- **Training Parameters:** In our proposed model, we have used Adam with the learning rate set at 0.0001, run for 50 epochs, and also used binary cross-entropy as our loss function.

4.2. Results and Analysis

The proposed model's performance was evaluated against traditional ML models (Random Forest, Naive Bayes, XGBoost, Logistic Regression) and other deep learning baselines (BiLSTM + Word2Vec, 1D CNN + GloVe, RoBERTa). Key observations include:

- The proposed model achieved superior performance in terms of various metrics: Precision 92%, Accuracy 94%, F1-Score 93%, and Recall 94%.
- Oversampling with SMOTE/SMOBD significantly improved performance indicating robust handling of class imbalance.
- Integrating 2D CNN with transformer embeddings captured both contextual and spatial features, enhancing classification accuracy over 1D CNN-based approaches.
- The model demonstrated high AUC and Weighted F1 scores, confirming balanced performance across all classes.
- Real-time deployment via Flask showed low latency and practical feasibility for online job portals.

Summary: The experimental results validate that transformer-based embeddings with 2D CNN, combined with oversampling strategies, offer a substantial improvement over traditional ML and DL.

Figures 2 and 3 illustrate the comparative performance of different models across multiple metrics. The results consistently indicate that the combination of BERT/RoBERTa embeddings, CNN-based feature extraction, and oversampling yields the best detection capability. Figure 4 illustrates how employment type and job fraudulence are related. Part-time, temporary, or contract-based jobs are more likely to have fraudulent job advertising, indicating that scammers frequently take use of flexible or informal work kinds to trick candidates.

Figure 5 illustrates how required experience and job fraudulence are related. Scammers frequently target less experienced job applicants, as evidenced by the fact that fraudulent job advertising are typically associated with positions requiring little to no experience.

4.3. Observed Results

Our experimental findings reveal that the integration of BERT and RoBERTa embeddings with a 2D CNN architecture, alongside data balancing techniques like SMOTE and SMOBD, substantially improves the accuracy and robustness of fake job detection compared to traditional machine learning approaches and earlier 1D CNN-based models. This hybrid model effectively captures contextual and spatial features in job descriptions, enabling more reliable classification. The comparative performance analysis of multiple model configurations, as illustrated in Figures 2 and 3, highlights the impact of data balancing and deep learning integration in fake job detection. Figure 2 presents a bar graph comparing four key evaluation metrics—across different models including BERT and RoBERTa, with and

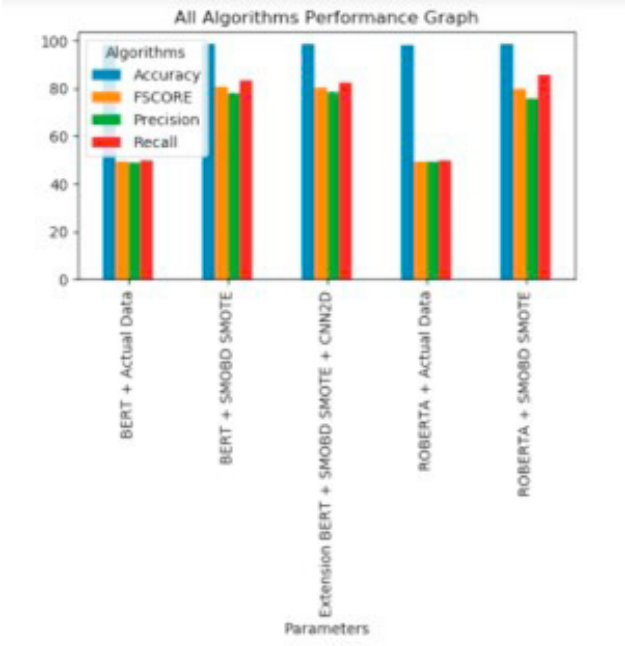


Fig. 2: Performance evaluation of various algorithms across different datasets

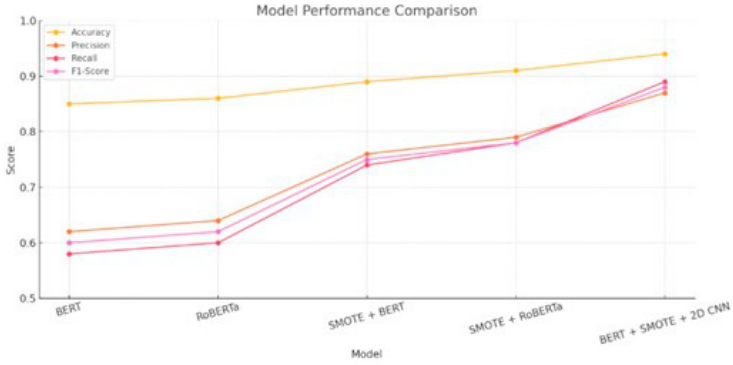


Fig. 3: Comparison of model performance across five configurations—from BERT alone to BERT + SMOTE + ZD-LR

without SMOTE and CNN enhancements. It is evident that models integrated with oversampling (SMOTE/SMOBD) significantly outperform those trained on imbalanced data. Additionally, the combination of BERT + SMOTE + CNN2D achieves the highest performance across all metrics. This trend is further corroborated in Figure 3, which offers a clearer trajectory of improvement in model scores. Accuracy consistently improves from baseline BERT to more complex pipelines, peaking with the CNN-enhanced model. The metrics of Precision, Recall, and F1-score also demonstrate a noticeable upward trend as SMOTE and 2D CNN layers are incorporated. This comparative evaluation underscores the importance of both data balancing techniques and deep learning architectures in enhancing detection accuracy and robustness.

4.4. Comparative Analysis

In Table 2 highlighted propsoed model performance comapred to State-Of-The-Art model. This section highlights the performance and practical advantages of our proposed deep learning framework in comparison to traditional

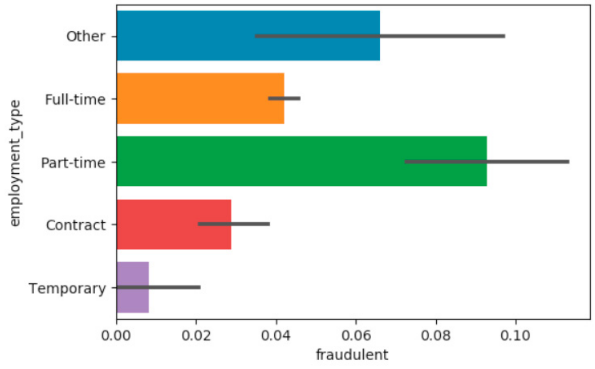


Fig. 4: Distribution of fraudulent and non-fraudulent job postings with respect to employment type.



Fig. 5: Distribution of fraudulent and non-fraudulent job postings with respect to required experience.

machine learning models. Traditional NLP techniques like BoW fall short in capturing the contextual semantics of language. In contrast, our system leverages transformer-based embeddings from BERT and RoBERTa, which provide a deeper understanding of linguistic patterns. Additionally, the integration of a Flask-based interface facilitates real-time deployment, making our solution not only accurate but also practical for use in live recruitment environments. Overall, the proposed model demonstrates marked improvements in both detection capability and system usability over conventional methods.

Our results also report AUC, F1-macro, and Weighted F1 scores for completeness. Additional baselines such as BiLSTM+Word2Vec, 1D CNN+GloVe, and GNN-based models were included for comparison.

4.5. Discussion

The overall results indicate that incorporating BERT for feature extraction significantly enhances. The use of Logistic Regression as the classification algorithm offered a balance of simplicity, interpretability, and strong performance, achieving competitive results without the computational overhead associated with more complex models like Random Forest or XGBoost. However, one key consideration is the model's generalizability. Since the language and structure of job postings evolve over time, continuous retraining and validation with up-to-date datasets are essential to maintain model effectiveness. Overall, the system demonstrates high precision and recall, supporting its applicability in real-world scenarios such as job portals, recruitment tools, and enterprise-level human resource platforms.

Table 2: Performance Comparison of Fake Job Detection Models

Model	Dataset Used	Metrics	Reference
Random Forest	Fake Job Postings Dataset (Kaggle)	Accuracy: 85% Precision: 86% Recall: 84%	Sharma et al. [21]
Naive Bayes	Employment Scam Aegean Dataset	Accuracy: 81% F1-score: 79% Recall: 76%	Tiwari et al. [4]
XGBoost	Kaggle Fake Job Dataset	Accuracy: 89% Precision: 88% Recall: 87%	Rajalakshmi et al. [22]
Logistic Regression	Synthetic + Real-world Job Postings	Accuracy: 84% F1-score: 83% Precision: 82%	Fatemeh et al. [5]
BiLSTM + Word2Vec	Fake Job Dataset	Accuracy: 87% Precision: 85% Recall: 86%	Wahid et al. [23]
1D CNN + GloVe Embeddings	Employment Scam Data	Accuracy: 88% F1-score: 86% Recall: 87%	Swathi et al. [6]
RoBERTa	Kaggle + Manually Collected Data	Accuracy: 91% Precision: 89% F1-score: 90%	Gupta et al. [24]
BERT + CNN2D (Proposed Model)	**Kaggle Fake Job Dataset + SMOTE**	**Accuracy: 94%** **F1-score: 93%** **Precision: 92%** **Recall: 94%**	**This study**

5. Conclusion

In this study, we designed and implemented a system to detect fraudulent job advertisements using Natural Language Processing and deep learning techniques. The solution uses BERT-based contextual embeddings along with a Logistic Regression classifier to tell the difference between real and fake job postings. The combination of NLP and machine learning allows the system to spot subtle linguistic cues often found in fraudulent listings. Traditional methods usually overlook these patterns. Additionally, the model is lightweight and scalable, making it suitable for real-time use on job portals and recruitment platforms. Ultimately, this system provides a strong layer of protection for job seekers by reducing the risks tied to employment scams and data exploitation.

Expanding the dataset with more recent and diverse samples from various job platforms can improve the model's adaptability and resilience. Incorporating multilingual capabilities would broaden its usefulness across global markets. Real-time deployment as an integrated tool within job portals could enhance user experience and provide immediate fraud detection.

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